Cosmological Stratigraphy: Recovering Pre-Indic Maritime Cosmology from Ritual Residues in Nusantaran Practice

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[DRAFT — Paper 5 of VOLCARCH Series — NOT FOR DISTRIBUTION]

> *Series context: This paper is the fifth in the VOLCARCH series. Papers 1-2 established taphonomic frameworks for volcanic and coastal invisibility of the archaeological record. Paper 3 developed a predictive settlement model. Paper 4 analyzed competitive inter-polity dynamics and the estuarine hybrid civilizational form. The present paper introduces a fourth axis of invisibility — cosmological taphonomy — arguing that imported religious frameworks systematically declassified indigenous maritime lifeways from the category of 'civilization,' producing an ideological burial layer that compounds the physical burial documented in earlier papers.*

Abstract

The invisibility of pre-Indic maritime civilization in Nusantara has been attributed in previous papers of this series to physical taphonomic processes: volcanic burial, coastal submersion, and organic decomposition. This paper identifies a fourth and qualitatively distinct mechanism of invisibility: cosmological taphonomy, the systematic displacement and reframing of indigenous knowledge systems by successive imported religious frameworks. We propose a subtraction methodology — Ajaran Asli = Contemporary Practice minus Imported Religious Layers — as a formal framework for isolating pre-Indic residues in current ritual, calendrical, cosmological, and linguistic practice. Three case studies are examined in depth: (1) the Javanese slametan mortuary ritual cycle (7-40-100-1000 days), whose numerical structure is incommensurable with any imported religion but maps precisely onto indigenous Javanese calendrical cycles; (2) the Pranata Mangsa ecological calendar, an indigenous astronomical system encoding millennia of maritime-agrarian phenological knowledge that survived Indianization, Islamization, and colonialism in active use; and (3) the La Galigo corpus of the Bugis, the world's longest known epic text, whose cosmology is explicitly and thoroughly maritime — representing the most complete surviving window into pre-Indic Nusantaran cosmological thought. We argue that these are not isolated survivals but fragments of a coherent and sophisticated indigenous knowledge system whose maritime orientation reflects the civilizational form documented in Paper 4. The paper concludes with a research agenda for systematic cosmological stratigraphy across Nusantara.

Keywords: cosmological taphonomy; Austronesian cosmology; slametan; Pranata Mangsa; La Galigo; ritual residues; maritime cosmology; Nusantara; cultural stratigraphy; indigenous knowledge

1. Introduction: The Fourth Layer of Invisibility

The VOLCARCH research program began with a straightforward empirical observation: Java's volcanic sedimentation buries archaeological sites at a measurable, quantifiable rate, rendering the material record of early Javanese civilization systematically invisible to surface

survey. Papers 1 through 4 have progressively expanded this core insight — from volcanic burial to coastal submersion to competitive institutional dynamics — building a cumulative framework for understanding why Nusantaran civilization is so much harder to see than its historical significance warrants.

This paper identifies a fourth and qualitatively distinct mechanism of invisibility, one that operates not on material culture but on the conceptual frameworks through which material culture is interpreted. We term this mechanism cosmological taphonomy: the process by which imported religious and cosmological systems displace, reframe, and effectively bury indigenous knowledge systems, causing them to become invisible not because they have been physically destroyed but because the interpretive frameworks that would allow them to be recognized have been replaced.

The argument is straightforward in its formal structure. Every major religion that entered Nusantara — Hindu-Buddhism, Islam, Christianity, and others — arrived from civilizational contexts with their own geographic character, and brought with them cosmological templates that reflected those contexts. Hindu-Buddhism originated in the river plains of northern India; its sacred geography is vertical and inland, organized around mountain cosmology and river ritual. Islam originated in the Arabian Peninsula; its sacred geography is desert and caravan, organized around pilgrimage to fixed points and the text-based community of believers. Neither template had obvious accommodation for the maritime, estuarine, and archipelagic character of Nusantaran civilization as documented in Paper 4.

The result was not destruction but displacement. Indigenous cosmological frameworks were not erased — they were overlaid, reinterpreted, and in many cases deliberately preserved in altered form by communities that recognized their practical value even as their religious content was nominally replaced. What survives is a set of ritual, calendrical, linguistic, and cosmological residues that the subtraction methodology proposed here can partially recover.

2. The Subtraction Methodology: Formal Framework

2.1 Cultural stratigraphy as analytical method

The geological concept of stratigraphy — reading time through the sequential layering of deposits — has a well-established analog in cultural anthropology. Cultural stratigraphy identifies successive layers of cultural influence and attempts to recover earlier layers from residues visible in the present. The methodology has been applied productively to language (historical linguistics reconstructing proto-languages from daughter language comparisons), to material culture (ceramic typology tracing exchange networks across millennia), and to mythology (comparative mythological analysis recovering proto-mythological narrative structures).

The present paper applies cultural stratigraphy to cosmological and ritual practice, using the following formal framework:

> *Ajaran Asli Nusantara = Contemporary Practice − (Hindu-Buddhist Layer + Islamic Layer + Christian Layer + Other Imported Layers) − Colonial Reinterpretation + Comparative Austronesian Baseline + Linguistic Fossil Record*

The subtraction terms represent documented influences that can be sourced to specific imported traditions. The additive terms — comparative Austronesian baseline and linguistic fossil record — provide external validation: if a residue identified by subtraction corresponds to patterns found in other Austronesian cultures that were not subject to the same imported influences (Taiwan aboriginal cultures, pre-Christian Polynesia, eastern Indonesia), this convergent evidence strengthens the inference that the residue is genuinely indigenous.

2.2 Methodological cautions

The subtraction methodology requires significant caution. First, the imported layers are not monolithic: 'Islam in Nusantara' is not a single uniform influence but a complex of Sufi, Shafi'i, and local syncretic variants that entered through different routes at different times. Second, the boundary between 'imported' and 'indigenous' is not always clear, as indigenous practices may have been modified to resemble imported ones while retaining indigenous content, and vice versa. Third, the comparative Austronesian baseline is itself reconstructed and carries its own inferential uncertainties.

These cautions do not invalidate the methodology but define its appropriate scope: it generates hypotheses about pre-Indic practice that require independent verification from linguistic, archaeological, and comparative ethnographic sources. The three case studies that follow are presented as demonstrations of the methodology's potential, not as definitive conclusions.

3. Case Study I: The Slametan Mortuary Cycle

3.1 The ritual and its anomalous structure

The slametan is a communal ritual feast found throughout Java and much of the Indonesian archipelago, performed on a wide range of occasions. Its mortuary variant — the commemoration of death at intervals of 7, 40, 100, and 1000 days after the event — is among the most widespread and deeply embedded ritual practices in Javanese culture, observed across religious boundaries by Muslims, Christians, and practitioners of Javanese traditional religion (Kejawen) alike.

The anomaly is immediately apparent when the numerical structure is compared against the mortuary traditions of the imported religions. Islam prescribes no specific commemorative intervals; the dominant Shafi'i tradition observed in Java explicitly discourages elaborate mortuary commemoration as potentially bid'ah (impermissible innovation). Hindu shraddha rituals observe the 13th day, monthly intervals, and annual commemorations — none matching the Javanese sequence. Tibetan Buddhist traditions observe a 49-day (7×7) bardo period, but Tibetan Buddhism had negligible presence in Java. The 7-40-100-1000 sequence is, in sum, incommensurable with any documented imported religious tradition.

3.2 Mapping the sequence onto indigenous calendrical systems

The Javanese calendrical system, documented in pre-colonial manuscripts and still partially in use, comprises several simultaneous cycles of which the most relevant here is the Pancawara or five-day market week (Pon, Wage, Kliwon, Legi, Pahing), which runs concurrently with the seven-day Saptawara week. The least common multiple of 5 and 7 is 35, meaning that a specific combination of Pancawara and Saptawara days recurs every 35 days — a cycle known as sepasar.

The mortuary intervals map onto this system with precision that strongly suggests intentional design. Seven days marks the completion of one Saptawara cycle — the first full week-cycle since death. Forty days (8 × 5) marks the completion of eight full Pancawara cycles — eight market weeks, a number with specific cosmological significance in Javanese numerology as a complete spatial unit (the eight cardinal and intercardinal directions). One hundred days (20 × 5) marks twenty Pancawara cycles — twenty being the number of human digits, representing the complete human being in Austronesian body-count numerology attested from Taiwan to Polynesia. One thousand days is approximately 200 Pancawara cycles and falls very close to the three-year mark — three full agricultural cycles in a Javanese volcanic

soil system, the period after which the deceased has been fully reintegrated into the ancestral layer.

The sequence is thus not arbitrary but precisely structured around indigenous time-reckoning systems that predate the entry of any imported religion. The Islamic vocabulary of the contemporary slametan (Quranic recitation, Arabic prayers) represents a surface layer imposed over a calendrical substrate of substantially greater antiquity.

3.3 The maritime interpretation

Beyond the calendrical structure, the slametan's underlying cosmological conception of death deserves analysis. The ritual premise is that the deceased undertakes a journey — the soul does not immediately depart but travels progressively from the world of the living to the world of ancestors, requiring communal provisioning and accompaniment at intervals along the route. This conception is structurally maritime: death as voyage, community as sending-off party, the living as those who remain at port while the dead sail toward the horizon.

This interpretation is not merely metaphorical. Comparative Austronesian mortuary traditions across the Pacific and Indian Ocean consistently frame death as a sea voyage: in Polynesia, the dead travel across the ocean to Pulotu or Hawaiki; in Madagascar (settled by Austronesian voyagers ~2000 BP), the dead journey across the water to ancestral lands; in eastern Indonesian traditions minimally affected by Indianization or Islamization, boats are a pervasive mortuary symbol, used as coffins, represented in grave goods, and depicted in rock art. The Javanese slametan, stripped of its imported religious vocabulary, encodes the same maritime cosmology of death as a navigated journey.

4. Case Study II: Pranata Mangsa — An Indigenous Astronomical System

4.1 Description and structure

Pranata Mangsa is a traditional Javanese ecological calendar dividing the year into twelve named seasons of unequal length, totaling 365 days. Unlike the Islamic Hijri calendar (lunar, 354 days), the Hindu-Buddhist Saka calendar (lunisolar), or the colonial Gregorian calendar, Pranata Mangsa is calibrated neither to lunar phases nor to solar solstices and equinoxes but to locally observable phenological phenomena: the flowering of specific plants, the arrival of specific migratory birds, the spawning behavior of specific fish, the appearance of specific insect species, and the behavior of winds and ocean currents.

Each of the twelve mangsa (seasons) is named and characterized by a specific set of ecological markers. Mangsa Kasa (June 22 – August 1, 41 days) is marked by the drying of springs and the appearance of specific forest fruits; Mangsa Karo (August 2 – August 25, 23 days) by the flowering of teak trees; through to Mangsa Kasada (June 3 – June 21, 19 days) which closes the cycle with the onset of the dry southeast monsoon. The precision of these phenological markers, and their tight correspondence with actual ecological events in central Javanese volcanic landscapes, indicates a system developed through millennia of careful systematic observation.

4.2 The maritime dimension

Pranata Mangsa is conventionally described as an agricultural calendar — a system for timing rice cultivation. This description is accurate but incomplete. Several mangsa are

explicitly calibrated to maritime phenomena: ocean current shifts, fish spawning cycles, monsoon wind transitions, and coastal bird movements. These maritime markers are not incidental additions to an agricultural system but integral components, suggesting the calendar was developed by a population engaged in both farming and fishing — precisely the coastal-agricultural hybrid population that Papers 2 and 4 argue was the dominant civilizational form in pre-Indic Java.

Crucially, Pranata Mangsa is not derived from any imported astronomical tradition. Hindu-Buddhist Jyotisha astronomy, while highly sophisticated, is mathematically and observationally oriented toward planetary positions, lunar phases, and nakshatra (lunar mansion) transits — none of which feature in Pranata Mangsa. Islamic astronomical tradition (ilmu falak) is similarly lunar-solar in orientation. Colonial European astronomy brought the Gregorian calendar but had no interest in indigenous phenological systems. Pranata Mangsa survived not because any imported tradition endorsed it but because it was practically indispensable for anyone farming or fishing in central Java — its ecological accuracy made it impossible to replace.

4.3 Pranata Mangsa as encoded indigenous knowledge

The survival of Pranata Mangsa into the present, despite sustained pressure from imported calendrical systems over fifteen centuries, argues for its deep practical value and, by extension, for the depth of the observational tradition that generated it. A calendar this precisely calibrated to local ecological reality requires many generations of systematic observation, recording, and refinement — the kind of knowledge infrastructure that, in other civilizational contexts, we would unhesitatingly identify as scientific practice.

That this knowledge system was never documented in stone inscriptions, was never the subject of scholarly Sanskrit or Arabic commentary, and was never incorporated into the religious architecture of the Indianized kingdoms tells us something important: it belonged to the practical knowledge tradition of ordinary farmers and fishermen, not to the elite ritual specialists whose activities generated the monumental and textual record. It is, in the terminology of this series, taphonomically invisible by class as well as by cosmology.

5. Case Study III: La Galigo — The Maritime Cosmological Text

5.1 The text and its significance

The La Galigo is an epic narrative of the Bugis people of South Sulawesi, preserved in Lontara script manuscripts distributed across libraries in the Netherlands, Indonesia, and Malaysia. At an estimated 300,000 lines of verse — roughly four times the length of the Mahabharata — it is among the longest literary works in human history. It describes the creation of the world, the descent of divine beings to populate the earth, and the adventures of the culture hero Sawerigading across multiple generations.

La Galigo was not widely known in international scholarship until the twentieth century, when Dutch scholars began systematic collection and translation of manuscripts. Its obscurity relative to texts of comparable scale reflects precisely the cosmological taphonomy argued in this paper: it was written in a regional script with no Sanskrit or Arabic connection, it described a world in which Islam (which arrived in South Sulawesi in the early seventeenth century) had no place, and it belonged to a pre-Islamic ritual specialist tradition (the bissu, trans-gender religious officiants) that colonialism and Islamization had largely suppressed.

5.2 La Galigo's maritime cosmology

What makes La Galigo uniquely valuable for this paper is the character of its cosmology, which is explicitly, pervasively, and structurally maritime. The three-layered cosmos of La Galigo — the Upper World (Boting Langi), the Middle World (Ale Lino), and the Lower World (Buri Liu) — is vertically organized, which might suggest Hindu influence. But the vertical axis is not a mountain: it is an ocean. The Upper World is reached by sailing upward; the Lower World is an underwater realm; the connections between worlds are navigational rather than terrestrial.

The divine beings who descend to populate the earth do so by boat. The central narrative of Sawerigading is structured as a series of sea voyages to distant lands in search of the woman he loves. The most sacred objects in the text are not temples or mountains but ships. The measure of a ruler's greatness is not the size of his temple complex but the quality of his fleet and the extent of his navigational knowledge. Death, as in the slametan analysis above, is framed as a sea voyage to the Lower World realm.

This is not the cosmology of a civilization that arrived at the sea as a secondary adaptation. This is the cosmology of a civilization for which the sea was the primary category of existence — the medium through which the universe is structured, through which power is expressed, through which the sacred is accessed, and through which life transitions (birth, marriage, death) are conceptualized.

5.3 La Galigo as a window into the pre-Indic baseline

South Sulawesi's Islamization occurred relatively late (early seventeenth century) and the Hindu-Buddhist influence in the region, while present, was less transformative than in Java or Bali. This means La Galigo preserves a cosmological tradition at a less processed stage than comparable Javanese or Balinese material — closer to the pre-Indic Austronesian baseline that the subtraction methodology attempts to reconstruct.

The parallels between La Galigo's maritime cosmology and the cosmological residues identified in the slametan (death as sea voyage) and Pranata Mangsa (sea-calibrated ecological calendar) are not coincidental. They are fragments of the same underlying tradition — the cosmological system of the estuarine hybrid civilization documented in Paper 4, expressed in different media and preserved to different degrees by different mechanisms, but recognizably coherent when read together.

6. Additional Residues: A Survey

6.1 Architectural orientations: kaja-kelod in Bali

Balinese sacred spatial orientation operates on a kaja-kelod axis: kaja (toward the mountain) is the direction of the sacred, purity, and the gods; kelod (toward the sea) is the direction of the impure, the dangerous, and the dead. This might appear to be a straightforward Hinduization — the mountain is sacred, the sea is profane. But the analysis is more complex. In Hindu cosmology as practiced in India, the sea is simply absent from the primary sacred geography — it is neither sacred nor profane but simply irrelevant. In Bali, the sea is profane but present — it is a direction, it has cosmological weight, it is not ignored. This retention of the sea as a cosmologically significant axis, even when revalued as negative, preserves a structural memory of a cosmological system in which the sea was the primary spatial reference.

6.2 Wayang: shamanic substrate beneath Hindu narrative

The Javanese wayang (shadow puppet theatre) uses narratives drawn directly from the Hindu epics — Mahabharata and Ramayana — and is thus superficially a product of Indianization. But the functional structure of the wayang performance is not Hindu. The dalang (puppeteer) functions as a mediator between the world of the living and the world of spirits and ancestors — a shamanic role with no equivalent in Hindu ritual specialist traditions. The opening and closing rituals of a wayang performance invoke local spirits and ancestral forces, not Hindu deities. The gunungan (mountain-tree-ocean composite figure used as a curtain between scenes) encodes a cosmological structure — mountain, tree, and ocean as three axes of the universe — that is specifically Austronesian rather than Indic. The Hindu stories are the surface; the shamanic ritual structure beneath them is the indigenous substrate.

6.3 Danyang and site-specific ancestral spirits

The widespread Javanese belief in danyang — ancestral spirits attached to specific locations, particularly large trees, springs, crossroads, and prominent landscape features — has no direct counterpart in any imported religion. Islamic tradition acknowledges jin (spirits) but these are not site-specific in the danyang sense and are not ancestral. Hindu tradition has various nature spirits but the specific form of the danyang as a named, localized ancestor-spirit accessible to the community through specific ritual protocols is distinctively Javanese. The danyang tradition encodes a conception of landscape as relationally inhabited — every significant feature of the environment is occupied by a presence with whom the community maintains an ongoing social relationship. This is not animism in the pejorative sense but a sophisticated relational ontology that constitutes the indigenous Nusantaran equivalent of what Western philosophy calls the philosophy of nature.

6.4 Linguistic fossils

Language preserves cosmological residues that are resistant to conscious modification because they are embedded in everyday vocabulary rather than in explicit ritual practice. Several Javanese and Malay lexical fields retain structures that reflect pre-Indic maritime cosmology. The Javanese term for 'east' — wetan — is etymologically connected to the direction from which trade winds bring rain across the Java Sea, not to the direction of sunrise as in the Sanskrit-derived terminology used in formal Hindu-Buddhist spatial orientation. The term for 'north' — lor — has no Sanskrit cognate and appears to derive from a proto-Austronesian navigational term. The vocabulary of social relationships in Javanese and Malay is extensively organized around maritime metaphors: one 'anchors' a relationship, 'navigates' social situations, follows the 'current' of social consensus — a fossil record of a society that thought about the social world in nautical terms.

7. Toward a Systematic Cosmological Stratigraphy

7.1 Research agenda

The three case studies and survey of additional residues presented above demonstrate both the potential and the complexity of the cosmological stratigraphy methodology. A systematic program would require collaboration across disciplines currently operating in relative isolation: ritual studies and religious anthropology, comparative Austronesian linguistics, archaeoastronomy and ethnoastronomy, comparative mythology, and the taphonomic archaeological framework developed in this series.

Priority research questions include: (1) systematic mapping of ritual practices across Nusantara that cannot be sourced to imported religious traditions, using the subtraction

methodology formally; (2) computational analysis of the La Galigo corpus for structural patterns connecting it to other Austronesian epic traditions; (3) comparative study of Pranata Mangsa-type ecological calendars across the archipelago, assessing geographic distribution and local variation; (4) linguistic analysis of cosmological vocabulary in Austronesian languages of eastern Indonesia, minimally affected by Indianization and Islamization, to reconstruct the pre-Indic semantic field for key cosmological concepts.

7.2 The political and ethical dimensions

Cosmological stratigraphy in an Indonesian context carries political and ethical dimensions that must be acknowledged explicitly. Indonesia is the world's largest Muslim-majority nation, and scholarly analysis that frames Islam as a layer overlying an indigenous substrate may be read as implying a hierarchy of authenticity — indigenous good, imported bad — that is both analytically false and politically irresponsible. This paper explicitly rejects such a reading.

The argument is not that imported religions diminished Nusantaran civilization. The argument is that the cosmological frameworks of imported religions — arising from geographically specific contexts — did not have natural categories for the maritime civilizational form that was indigenous to Nusantara, and that this categorical mismatch contributed to the historical invisibility of that civilizational form in both the scholarly and popular record. Recovering the indigenous layer is not about privileging it over imported layers but about achieving a more complete and accurate picture of the full depth of Nusantaran civilizational history.

7.3 The recursive connection to the VOLCARCH framework

The cosmological taphonomy argument connects recursively to the physical taphonomy documented in earlier papers of this series. Papers 1 and 2 argued that the physical record of maritime civilization in Nusantara is systematically invisible because of volcanic burial and coastal erosion. Paper 4 argued that the institutional record of estuarine hybrid civilization is underrecognized because of historiographic bias toward monumental and textual evidence. The present paper argues that the cosmological record of pre-Indic maritime civilization is partially recoverable from ritual residues, but that its recovery requires recognizing the displacement effect of imported cosmological frameworks.

Together these four mechanisms — volcanic burial, coastal erosion, historiographic bias, and cosmological displacement — constitute a comprehensive explanation for why Nusantaran civilization appears, from the conventional scholarly record, so much thinner and less complex than the convergent evidence of archaeology, genetics, linguistics, and comparative ethnography suggests it actually was. The VOLCARCH program is ultimately an argument for recovering the full depth of that complexity — layer by layer, from sediment and sea, from institutional dynamics and cosmological residues, until a more complete picture of the human past in one of the world's most extraordinary civilizational zones becomes visible.

8. Conclusions

Nusantaran civilization has been rendered invisible by four successive mechanisms documented across this paper series: volcanic sedimentation burying its material remains, sea-level rise submerging its coastal infrastructure, historiographic bias erasing its institutional complexity, and cosmological displacement obscuring its indigenous knowledge systems. This paper has addressed the fourth mechanism, demonstrating through three

case studies that significant residues of pre-Indic maritime cosmology survive in contemporary Nusantaran practice.

The slametan mortuary cycle preserves an indigenous Javanese calendrical substrate and a maritime conception of death as sea voyage. The Pranata Mangsa encodes millennia of maritime-agrarian ecological observation in an astronomical system incommensurable with any imported tradition. La Galigo preserves the most complete surviving expression of pre-Indic Austronesian maritime cosmology, in which the sea is the primary medium of existence, power, and sacred connection.

These are not isolated curiosities. They are fragments of a coherent cosmological tradition that once structured the conceptual world of the estuarine hybrid civilizations documented in Paper 4 — civilizations whose material record has been erased by the physical taphonomic processes documented in Papers 1 and 2. Recovering these fragments through systematic cosmological stratigraphy is not an exercise in nostalgia or cultural nationalism. It is an exercise in intellectual honesty about the full depth of human civilizational achievement in one of the world's most remarkable archipelagos.

Draft Notes & Placeholders

[ ] Section 3.2 — The Pancawara/Saptawara calendrical analysis needs verification against primary Javanese manuscript sources. Check Zoetmulder (1982) Old Javanese-English Dictionary and Ricklefs Javanese calendar studies.

[ ] Section 3.3 — The maritime death voyage comparative evidence needs ethnographic citation. Check Fox (1980) The Flow of Life: Essays on Eastern Indonesia and Bellwood (1997) Prehistory of the Indo-Malaysian Archipelago.

[ ] Section 4.1 — Pranata Mangsa description should be verified against BMKG (Indonesian Meteorological Agency) documentation and Daldjoeni (1984) Pranatamangsa.

[ ] Section 5.1 — La Galigo manuscript locations: Universiteitsbibliotheek Leiden (largest collection), KITLV, Makassar National Archive. Noorduyn (1955) and Pelras (1996) The Bugis are key secondary sources.

[ ] Section 5.2 — The three-layered cosmic ocean structure of La Galigo needs closer textual analysis — this claim should be verified directly against translated passages.

[ ] Section 6.4 — Linguistic fossil claims (wetan etymology, etc.) need verification from Austronesian comparative linguistics. Check Blust (2013) The Austronesian Languages.

[ ] Section 7.2 — The political/ethical section is essential but needs careful framing. Consider consultation with Indonesian colleague before submission.

[ ] Journal target: this paper is most interdisciplinary of the series — consider Indonesia and the Malay World, Bijdragen tot de Taal-, Land- en Volkenkunde (BKI), or Journal of the Royal Anthropological Institute.

[ ] Consider whether La Galigo deserves its own standalone paper — the corpus is large enough and significant enough to sustain one.

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[Additional references to be added upon full draft completion]